

Review: Slope as rate of change



1 Determine whether the following is a correct description of slope:

a $\frac{\text{Horizontal rise}}{\text{Vertical run}}$

b $\frac{\text{Vertical rise}}{\text{Horizontal run}}$

c $\frac{\text{Horizontal run}}{\text{Vertical rise}}$

d $\frac{\text{Vertical run}}{\text{Horizontal rise}}$

2 State the term used to the change in the horizontal distance along a line.

3 State the definition of the slope of a line.

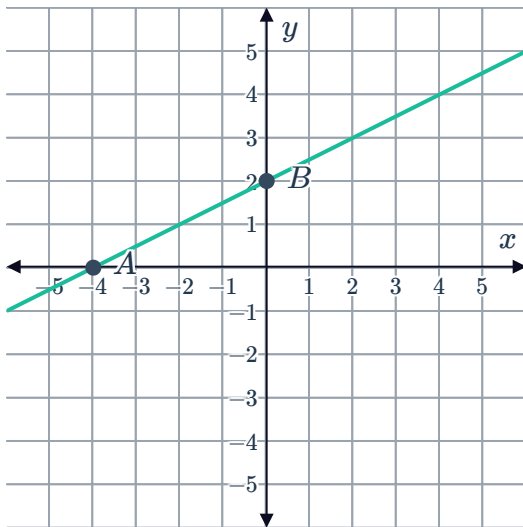
4 For each of the following segments containing A and B :

i Find the rise going from A and B .

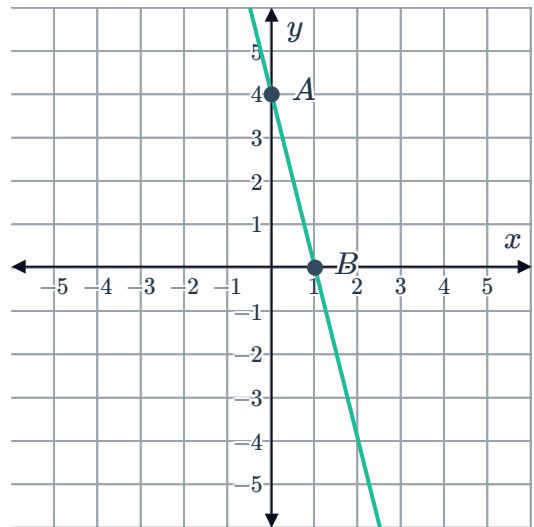
ii Find the run going from A and B .

iii Find the slope of the $\overline{AB}$.

a $A(-4, 0)$ and $B(0, 2)$



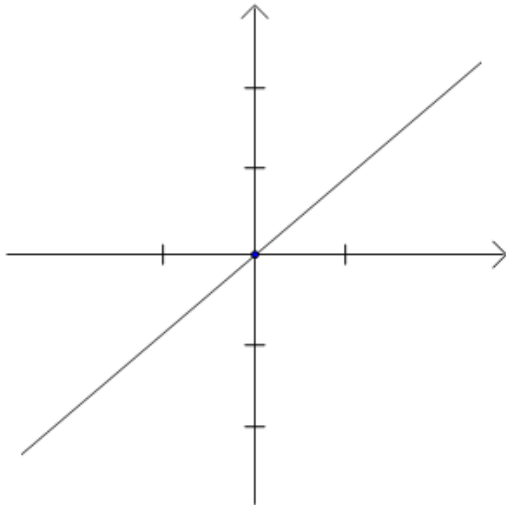
b $A(0, 4)$ and $B(1, 0)$



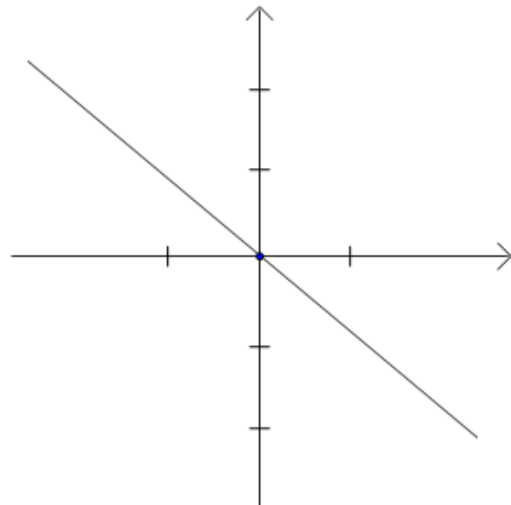
5 Identify what kind of slope the following line has.



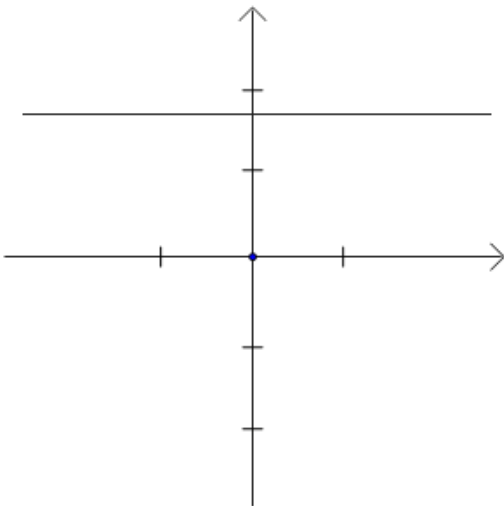
a



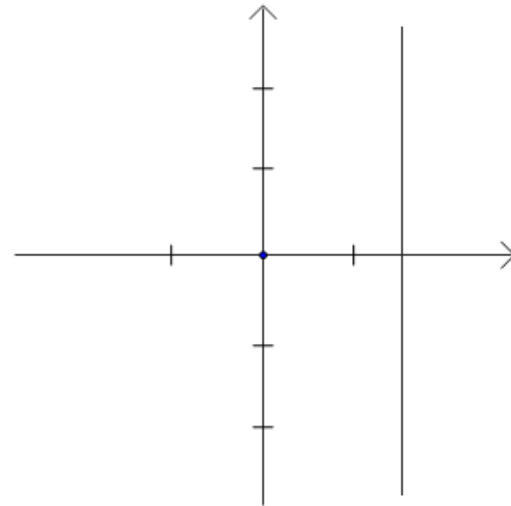
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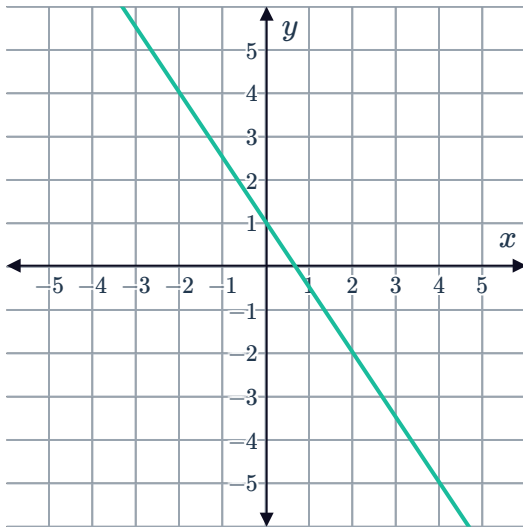


6 A line crosses the x -axis at 7, and crosses the y -axis at -6 .
Is the slope of the line positive or negative?

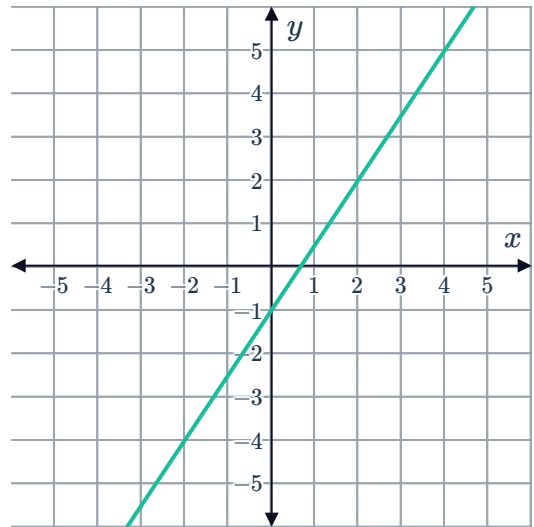
- 7 The slope of a line is $\frac{3}{2}$, and when $x = 0$, $y =$ 

Determine whether the following could be the graph of the line:

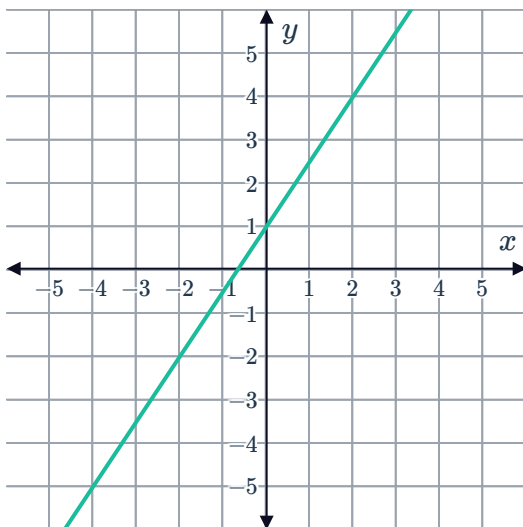
a



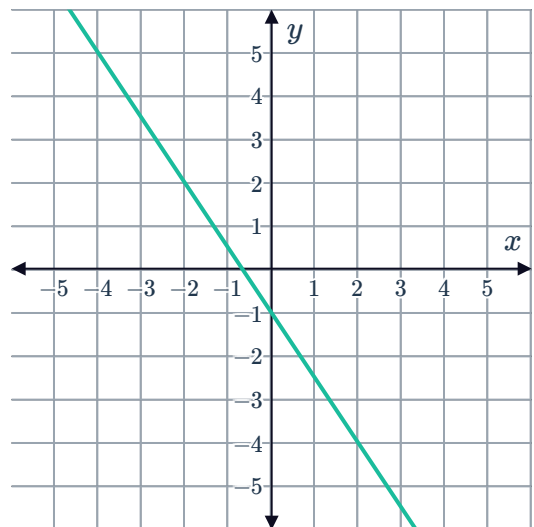
b



c



d



- 8 A staircase is to be built so that its maximum steepness is 0.6.
If each step goes in 22 cm, solve for the maximum height, in centimeters, it can rise vertically.
- 9 A paratrooper falls to the ground along a diagonal line. His fall begins 1157 m above the ground, and the line he follows has a slope of 1.3. That is, he falls 1.3 m vertically for every 1 m he moves across horizontally.
How far horizontally across the ground does he land from his initial position in the sky?

- 10 Gas costs a certain amount per gallon. The following table shows the cost in dollars of various amounts of gas.



Number of gallons (x)	0	10	20	30	40
Cost of gas (y)	0	12.70	25.40	38.10	50.80

- a Write an equation linking the number of gallons of gas pumped (x) and the cost of the gas (y).
 - b How much does gas cost per gallon?
 - c How much would 73 gallons of gas cost at this unit price?
 - d In the equation, $y = 1.27x$, what does 1.27 represent?
- 11 A diver starts at the surface of the water and begins to descend below the surface at a constant rate. The following table shows the depth of the diver over 5 minutes.

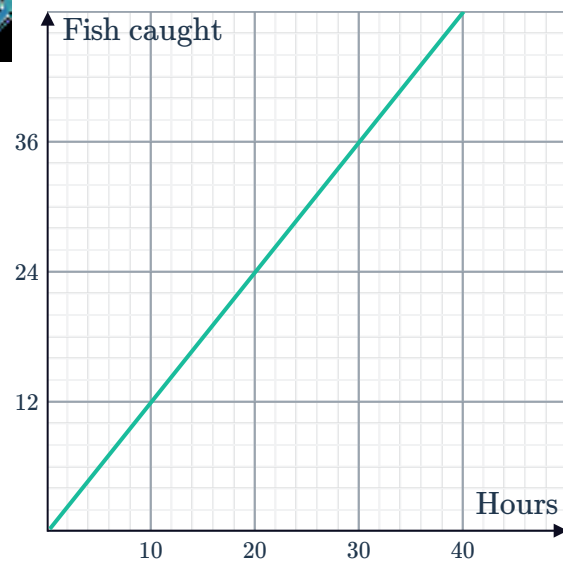
Number of minutes passed (x)	0	1	2	3	4
Depth of diver in meters (y)	0	1.4	2.8	4.2	5.6

- a What is the increase in depth each minute?
 - b Write an equation for the relationship between the number of minutes passed (x) and the depth (y) of the diver.
 - c In the equation, $y = 1.4x$, what does 1.4 represent?
 - d At what depth would the diver be after 6 minutes?
 - e How long does the diver take to reach 12.6 m beneath the surface?
- 12 A dam used to supply water to the neighboring town had the following data recorded for its volume over a number of months.

Month	1	2	3	4
Volume (billions of liters)	134	117	100	83

At what rate is the volume of water in the dam changing?

- 13 Consider the proportional relationship shown in the graph, showing the number of fish caught over time.



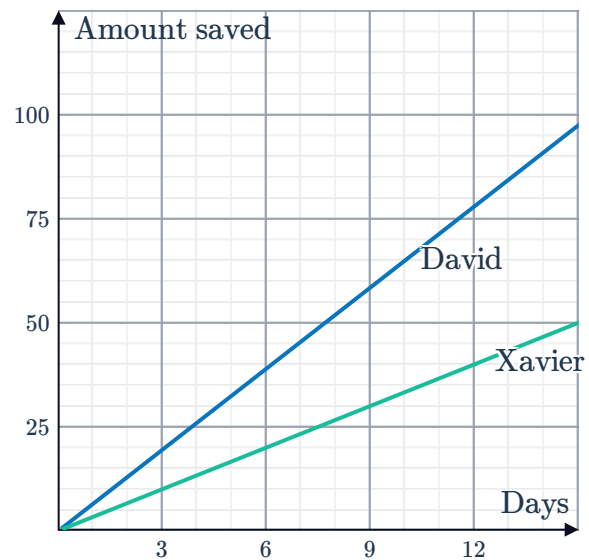
- What is the unit rate of this relationship in fraction?
- Interpret your answer from part (a).
- What feature of the straight line graph does the rate of change relate to?

- 14 David and Xavier are saving money for a vacation. David knows that he can represent his savings over time using the equation $y = \frac{13}{2}x$, where x represents the number of days and y represents the savings.

- Xavier saves \$10 every 3 days. Complete the table showing Xavier's savings over time.

Days	1	6	9	12
Dollars saved	10	20	30	1

- Comparing the graphs of each person's savings, how can you tell who is saving more per day?
- Using the graph, who has the greater rate of savings?



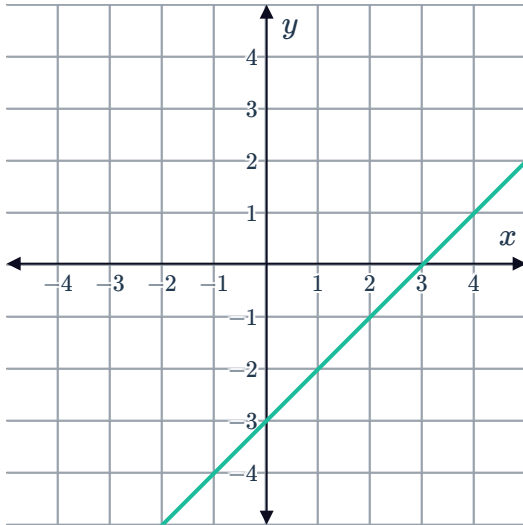


Additional questions

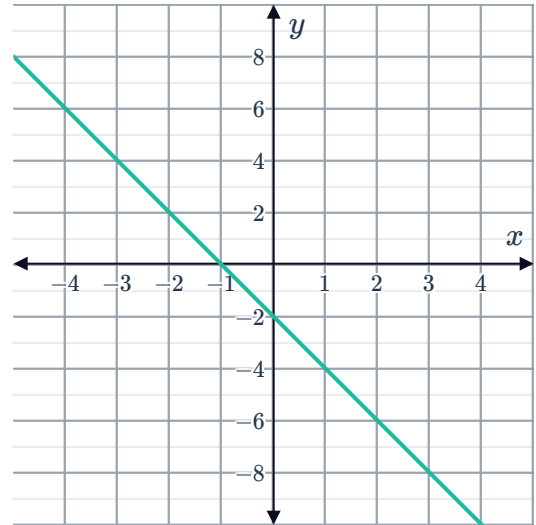
15 For the following functions:

- i State whether the function is increasing, decreasing, or constant.
- ii State the rate of change.

a



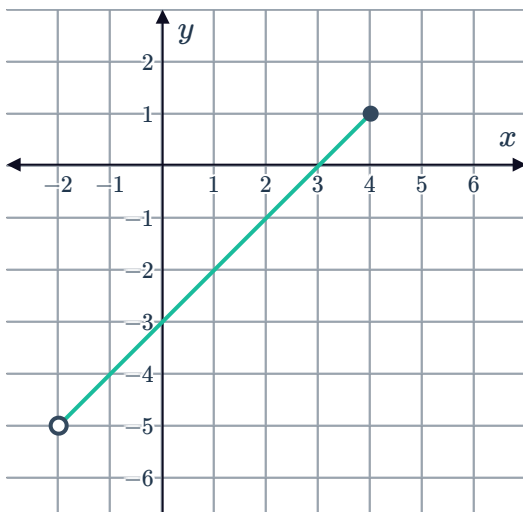
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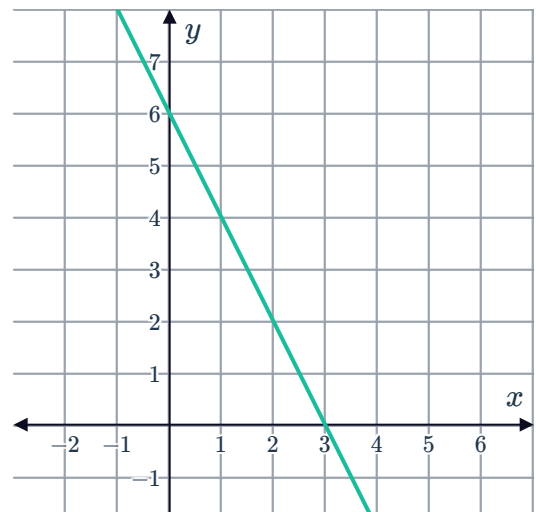
16 For each of the following graphs:

- i State the domain.
- ii State the range.
- iii State the x - and y -intercepts.
- iv State the rate of change.

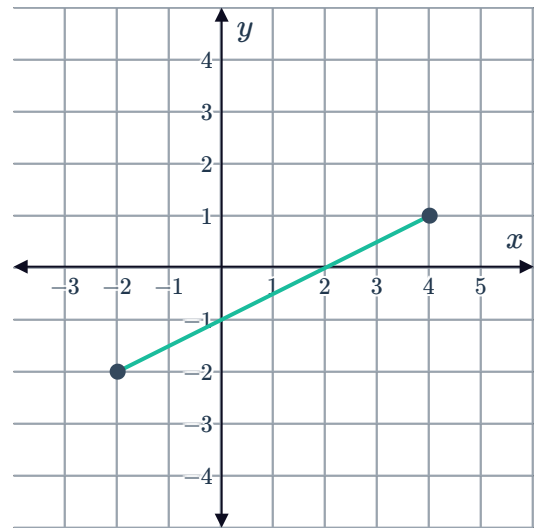
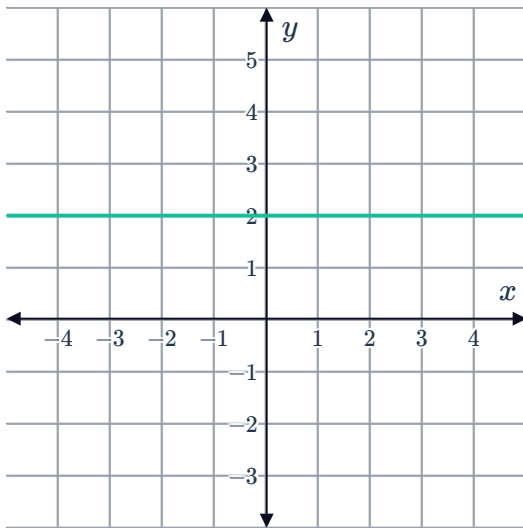
a



b

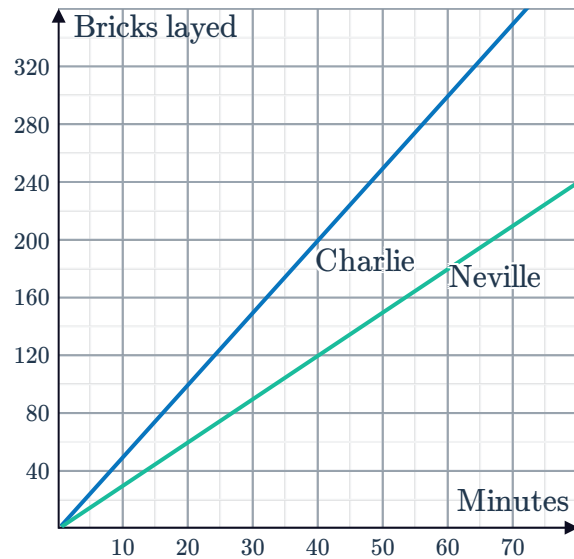


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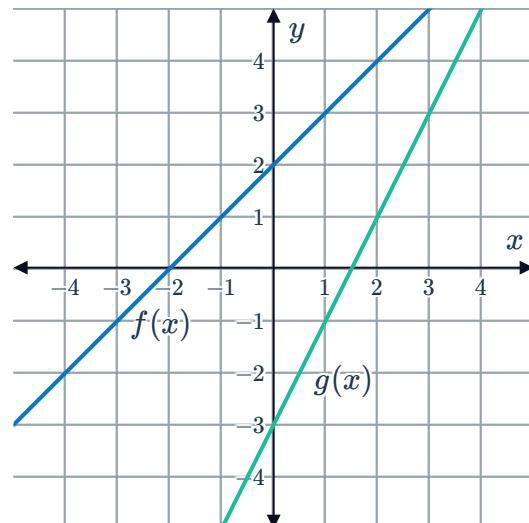


17 Two construction workers are competing to see who can lay the most bricks in one hour. The graph shows the number of bricks laid and the time, in minutes.

- a Interpret the rate of change of the given lines.
- b Determine who can lay more bricks in 60 minutes.
- c Explain why the y -intercept of both lines is 0.



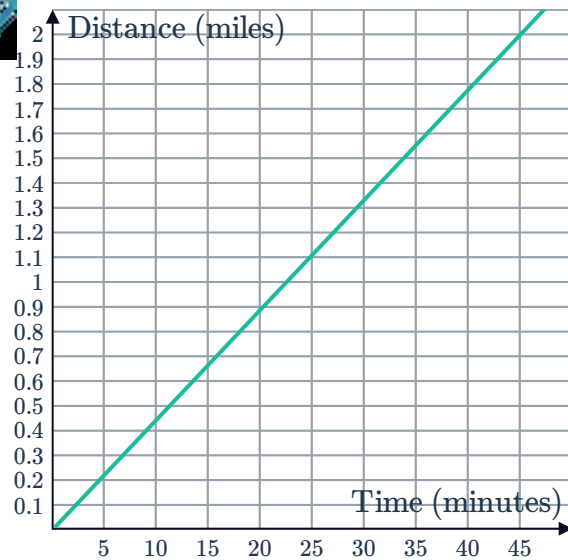
18 Alicia has concluded that the rate of change is greater in $f(x)$ than $g(x)$ because it is higher on the given graph. Explain and correct her error.



19 The following graph shows the distance Natalia swam during a recent open water swim:



- a What is the slope of the line?
- b What does the slope of the line represent in terms of the situation?



20 The following graph shows the number of batches of cookies produced in a factory:

- a What is the slope of the line?
- b What does the slope of the line represent in terms of the situation?

